Analytics Vidya Course Search Engine Report

Project Files: https://github.com/Satwik-uppada/Analytics-Vidya/tree/main

1. model.ipynb: This file handles data preparation and model development.

2. app.py: This file implements the Streamlit web application for the search engine.

3. **analytics_vidhya_courses.csv:** This file contains Course name, image url, course url, lessons, price.

4. **course_details.csv:** This file contains course name, description, curriculum.

Data Preparation and Model Development (model.ipynb)

The model.ipynb file performs the following steps:

- **Data Loading:** Loads course data from two CSV files:
  - analytics_vidhya_courses.csv: Contains basic course information like title, lessons, price, image URL, and course link.
  - course_details.csv: Contains detailed course information like description and curriculum.
- **Data Cleaning:**
  - Filters the data to include only free courses.
  - Handles missing values in the Lessons column by filling them with an empty string.
- **Data Merging:** Merges the two dataframes based on the Course Name column.
- **Curriculum Flattening:** Converts the dictionary-like structure of the Curriculum column into a readable string.
- **Combined Text Generation:** Creates a new column "Combined Text" by concatenating the Course Name, Description, and Curriculum columns.
- **Embedding Generation:** Uses the **"all-mpnet-base-v2" SentenceTransformer model** to generate embedding vectors for each course based on the Combined Text.

```python
from sentence_transformers import SentenceTransformer
model = SentenceTransformer('all-mpnet-base-v2')

final_data["Embedding Vector"] = final_data["Combined Text"].apply(lambda x: model.encode(x))
```

- **Data Saving:** Saves the processed data to a CSV file final_Data.csv.
- **Elasticsearch Indexing:**

  Elastic Search Cloud: search-tool - Elastic

  Elastic Search URL: https://my-elasticsearch-project-d00981.es.us-east-1.aws.elastic.cloud:443

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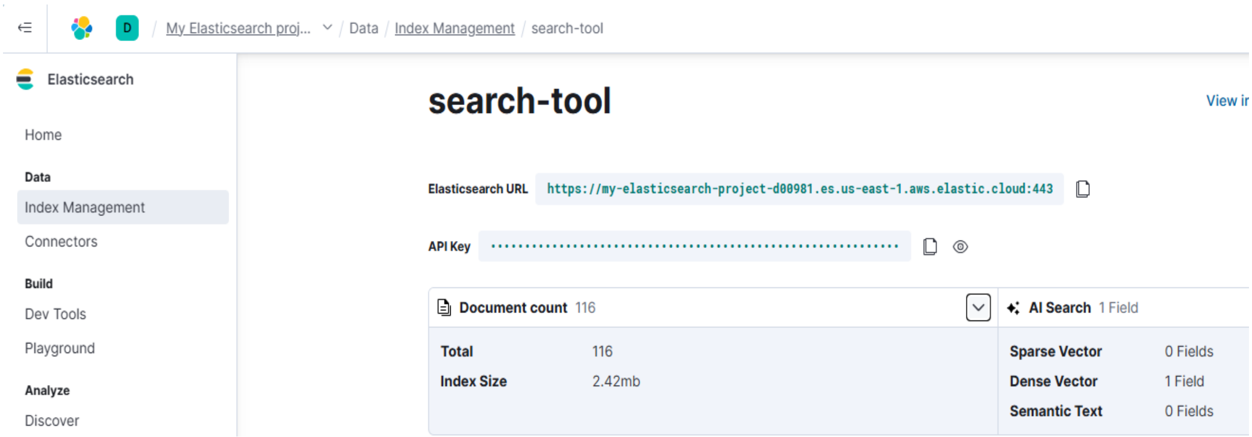
- Creates an Elasticsearch index named search-tool with appropriate mappings for each column.



- Indexes the course data into the Elasticsearch index.

Streamlit Web Application (app.py)

The app.py file implements the Streamlit web application for the search engine. It provides the following functionalities:

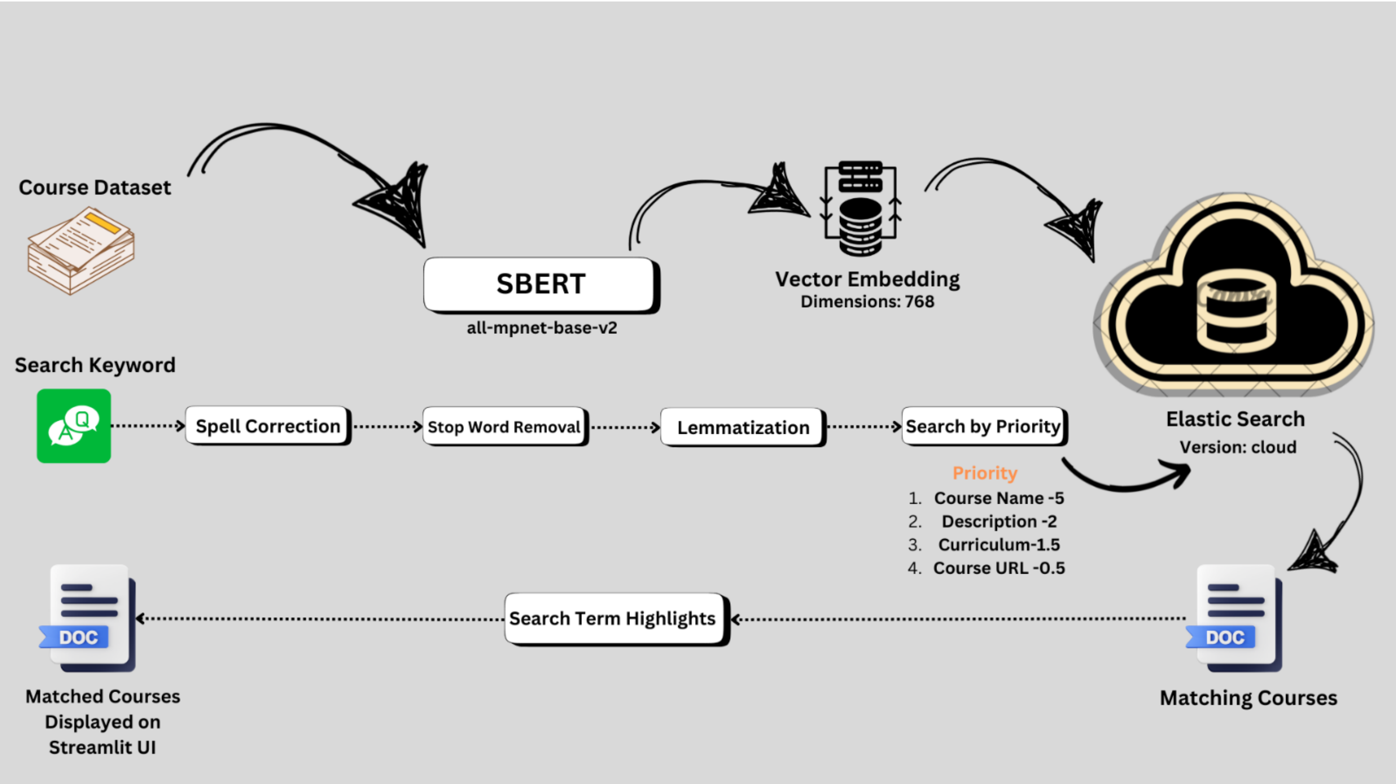
Huggingface Space: https://huggingface.co/spaces/Satwikuu/Smart_search_tool

- **User Interface:**
  - Displays a title "Course Search Engine" and a sidebar with a logo and search tips.
  - Provides a text input field for users to enter their search query.
  - Includes a sidebar checkbox to enable debug mode.
- **Search Functionality:**
  - Initializes an Elasticsearch client.
  - Corrects spelling mistakes in the user query using the correct_spelling function.
  - Performs a search using the search_courses function, which:
    - ✓ Splits the query into terms for highlighting.
    - ✓ Lemmatizes and removes stop words from the query.
    - ✓ Constructs a complex Elasticsearch query using bool and should clauses to prioritize matches in different fields.
    - ✓ Deduplicates results based on Course Link while keeping the highest scoring version.
    - ✓ Highlights search terms in the course name, description, and curriculum using HTML.
- **Result Display:**
  - Displays the search results in a card-like format.
  - Shows the course name, description, and curriculum (if available) with highlighted search terms.
  - Includes a "View Course Details" button that links to the course page.
  - Provides debug information (if debug mode is enabled) including search score, course URL, and total vs. unique matches.

Overall Analysis

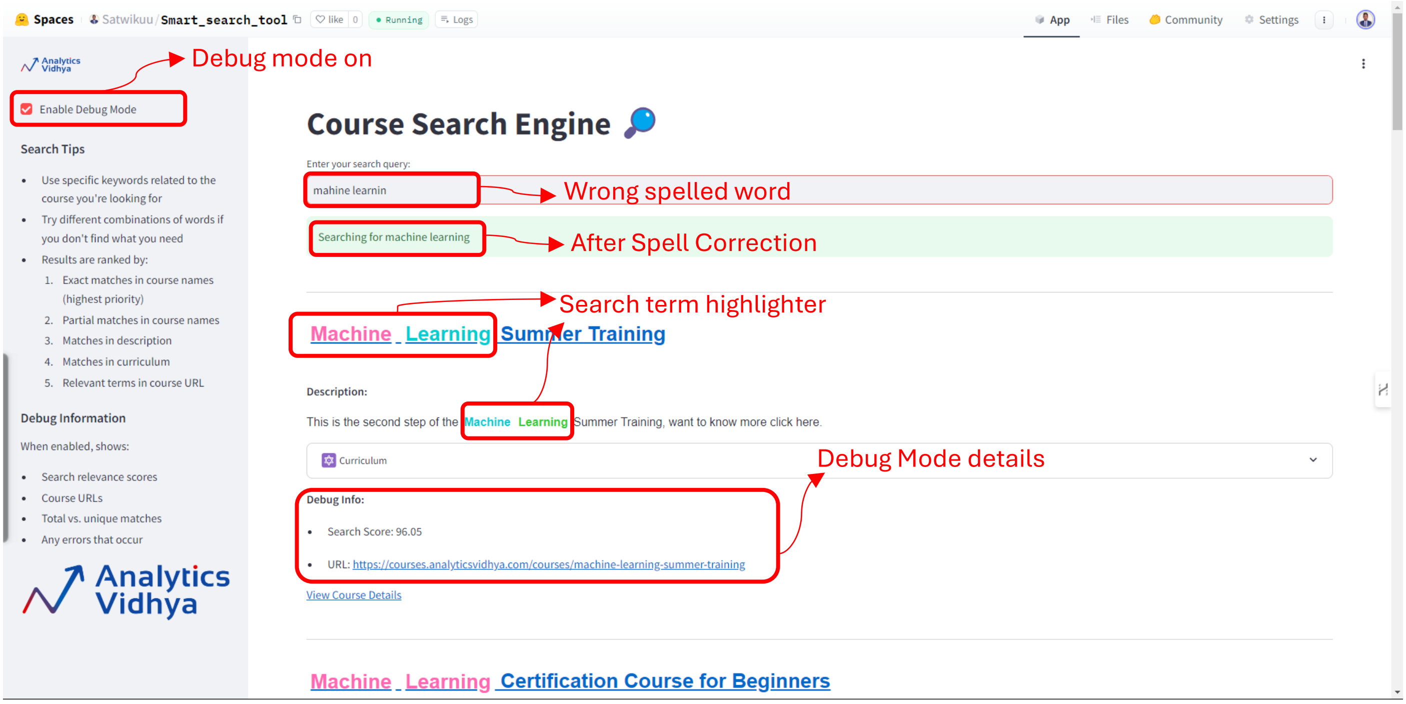
The Analytics Vidya Course Search Engine is a well-designed application that leverages the power of Elasticsearch and SentenceTransformers for efficient and accurate course search. The code is well-structured and includes features like spelling correction, query term highlighting, and result deduplication to enhance the user experience. The Streamlit web application provides a user-friendly interface for searching and browsing courses.

Process Flow Chart:



Advanced Semantic Search of Courses using SBERT and Elastic Search

Web App UI:



Reference:

1. SBERT: https://sbert.net/docs/sentence_transformer/pretrained_models.html
2. Elastic cloud: https://www.elastic.co/guide/en/cloud/current/ec-getting-started.html
3. https://medium.com/@abidsaudagar/build-semantic-search-with-elasticsearch-and-bert-vector-embeddings-from-scratch-41707cb4e2fd
4. https://medium.com/@abidsaudagar/building-advanced-semantic-search-with-elasticsearch-and-gpt-3-5-turbo-28647e155fda

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